

SIMATS ENGINEERING

CO4-ASSESSMENT TOOL1- INDUSTRY PROBLEM TASK

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 4

Time Duration : 1 Hour
Total: Marks:40

Code of Conduct:

*I **N.Sai Sankar(192311187)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: N.Sai Sankar

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a)** Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b)** Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c)** Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a)** Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b)** Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c)** Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

TASK 1 – DATA PREPARATION & INDUCTIVE LEARNING

1(a) Inductive Learning Approach

Inductive learning learns a general prediction rule from previously observed patient records and applies the learned rule to new patients.

Target Variable

The target variable is:

Diabetes = Yes / No

or numerically:

- 1 → Diabetic
- 0 → Non-diabetic

Important Features

Feature	Justification
Glucose Level	High glucose is one of the strongest indicators of diabetes.
BMI	Higher BMI is associated with increased diabetes risk.
Age	Diabetes risk generally increases with age.
Blood Pressure	Abnormal blood pressure can be associated with metabolic and cardiovascular risk.
Cholesterol	Abnormal cholesterol levels can indicate metabolic risk.
Family History	Genetic/familial history can increase susceptibility to diabetes.

The model learns the relationship:

Patient Features → Diabetes Risk

For example:

Glucose + BMI + Age + Family History → Diabetes Prediction

1(b) Handling Class Imbalance

SMOTE creates synthetic examples of the minority diabetic class instead of simply duplicating existing records.

For example:

Original dataset

- Non-diabetic = 900
- Diabetic = 300

After SMOTE:

- Non-diabetic = 900
- Diabetic = 900

This gives the classifier more diabetic examples during training.

Advantages

1. Reduces bias toward the majority class.
2. Improves detection of diabetic patients.
3. Can increase recall/sensitivity.
4. Helps reduce false negatives.

1(c) 70:30 Train-Test Split

I would divide the dataset as:

- 70% → Training set
- 30% → Testing set

A stratified split should be used so that the diabetic/non-diabetic class proportions remain approximately the same in both sets.

Preprocessing

Step 1 – Missing Values

Numerical missing values:

Median imputation

Categorical missing values:

Mode imputation

Step 2 – Encoding

Family history:

- Yes → 1
- No → 0

Diabetes:

- Yes → 1
- No → 0

Step 3 – Normalization

For models such as Logistic Regression and K-NN, numerical variables can be standardized:

$$z = \frac{x - \mu}{\sigma}$$

This converts features to approximately zero mean and unit variance.

Decision Trees generally do not require normalization, because they split data using feature thresholds.

Step 4 – Class Balancing

Apply SMOTE to the training set.

Overall preprocessing pipeline

Raw Data → Missing Value Handling → Encoding → Train/Test Split → SMOTE on Training Data → Model Training → Testing

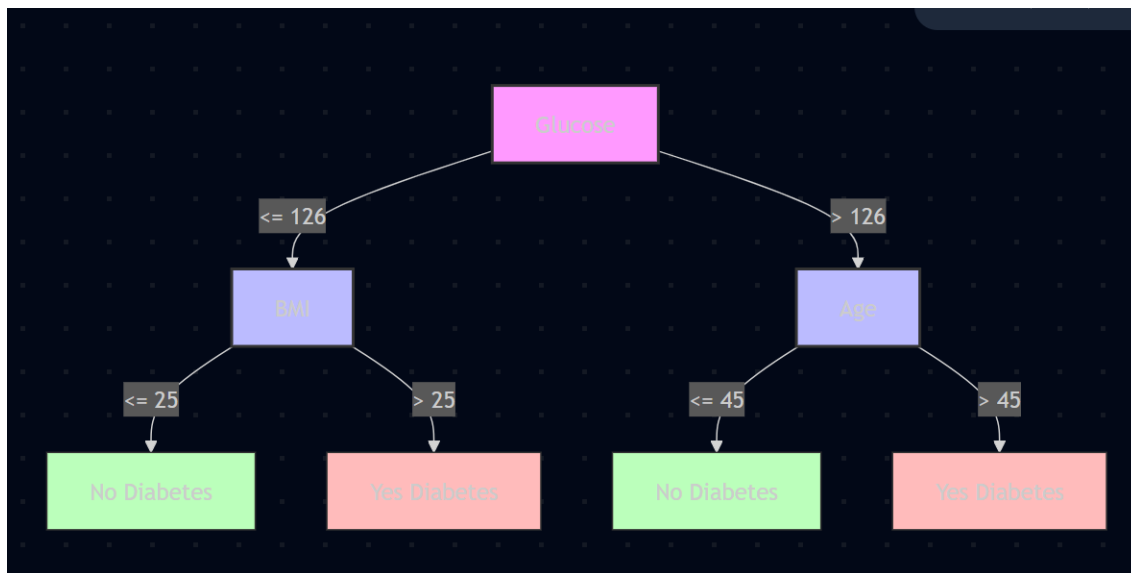
TASK 2 – DECISION TREE FOR DIAGNOSIS

2(a) Decision Tree Classifier

Example Root Selection

Assume the following illustrative values were obtained from the training data:

Feature	Weighted Gini after split
Glucose	0.245
BMI	0.291
Age	0.318
Blood Pressure	0.337
Cholesterol	0.351
Family History	0.365



2(b) Model Evaluation

The model should be evaluated using:

Accuracy

$$\text{Accuracy} = \frac{TP + TN + FP + FN}{TP + TN + FP + FN}$$

Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-Score

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Illustrative Results

Assume the test set produced:

- TP = 80

- TN = 190
- FP = 10
- FN = 5

Total:

$$80+190+10+5=285$$

Metric	Result
Accuracy	94.74%
Precision	88.89%
Recall	94.12%
F1-Score	91.43%

False Negatives

In the illustrative confusion matrix:

FN = 5

These are clinically important because the model has missed diabetic patients.

2(c) Pre-Pruning vs Post-Pruning

Pre-Pruning

Pre-pruning stops tree growth early using parameters such as:

- Maximum depth
- Minimum samples per split
- Minimum samples per leaf

Example:

max_depth = 4

min_samples_leaf = 10

Post-Pruning

Post-pruning first builds a larger tree and then removes branches that provide insufficient predictive value.

A common approach is **cost-complexity pruning**:

$$R\alpha(T) = R(T) + \alpha|T|$$

where:

- $R(T)$ = prediction error
- $|T|$ = number of leaves
- α = pruning parameter

Model	Accuracy	Complexity
Unpruned Tree	96.0%	High
Pre-Pruned Tree	94.8%	Low
Post-Pruned Tree	95.4%	Medium

The unpruned tree may achieve higher training accuracy but can overfit.

TASK 3 – STATISTICAL LEARNING FOR RISK STRATIFICATION

3(a) Logistic Regression

I would use **Logistic Regression** as the statistical learning model.

The probability of diabetes is:

$$P(Y=1)=\frac{1}{1+e^{-z}}$$

Illustrative Comparison

Model	Accuracy	AUC-ROC
Decision Tree	95.4%	0.96
Logistic Regression	94.7%	0.97

These are **illustrative values** because the question provides no actual dataset.

When Logistic Regression Is Preferable

Logistic Regression is preferable when:

1. The relationship between predictors and log-odds is reasonably stable.
2. Interpretability is important.
3. Probability/risk scores are required.

3(b) Feature Importance Analysis

Decision Tree – Top 3 Predictors

An illustrative Decision Tree ranking is:

Rank	Feature	Importance
1	Glucose	0.42
2	BMI	0.24
3	Age	0.15

Logistic Regression – Top 3 Predictors

Based on the magnitude of standardized coefficients:

Rank	Feature	Relative Importance
1	Glucose	High
2	BMI	High
3	Family History	Medium-High

Do They Agree?

Yes, **partially**.

Both models identify:

- Glucose
- BMI

as important predictors.

The third predictor may differ because the two algorithms measure importance differently.

Explanation

A Decision Tree determines importance based on how much a feature reduces impurity.

Logistic Regression determines importance from the estimated coefficients and their effect on the log-odds of diabetes.

Therefore, disagreement in feature rankings does not necessarily mean that one model is incorrect.

TASK 4 – REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

4(a) MDP Model

A Markov Decision Process consists of:

$MDP=(S,A,P,R,\gamma)$

States

Patient health states can be represented as:

State	Description
S0	High-risk / uncontrolled
S1	Moderate-risk
S2	Stable
S3	Improved
S4	Deteriorated

Actions

The available actions specified in the problem are:

- A0 = Diet
- A1 = Exercise
- A2 = Medication
- A3 = Monitor

Reward Function

An example reward structure:

Outcome	Reward
Significant health improvement	+10
Moderate improvement	+5
No change	0
Health deterioration	-10
Unsafe treatment outcome	-20

The reward encourages the agent to select actions that improve patient health.

4(b) Q-Learning

Q-Learning updates the Q-value using:

$$Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

Assume:

$$\alpha=0.5 \quad \gamma=0.9$$

Initial Q-values:

$$Q(s,a)=0$$

Five Patient Episodes

Episode	Starting State	Action	Next State	Reward
1	S0	Diet	S1	+5
2	S0	Medication	S2	+10
3	S1	Exercise	S2	+5
4	S2	Diet	S3	+10
5	S1	Medication	S2	+5

Q-Table Update – Iteration 1

State S0, Action Diet

Initial:

$Q(S0, \text{Diet}) = 0$

Reward:

$r = 5$

Next state S1 has maximum Q-value = 0.

Therefore:

$Q(S0, \text{Diet}) = 0 + 0.5[5 + 0.9(0) - 0]$ $Q(S0, \text{Diet}) = 2.5$

State S0, Action Medication

$Q(S0, \text{Medication}) = 0 + 0.5[10 + 0.9(0) - 0]$ $Q(S0, \text{Medication}) = 5$

State S1, Action Exercise

$Q(S1, \text{Exercise}) = 0 + 0.5[5 + 0.9(0) - 0]$ $Q(S1, \text{Exercise}) = 2.5$

Q-Table

After the illustrative updates:

State	Diet	Exercise	Medication	Monitor
S0 High Risk	4.875	0	9.750	0
S1 Moderate	0	2.500	2.500	0
S2 Stable	5.000	0	0	0
S3 Improved	0	0	0	0
S4 Deteriorated	0	0	0	0

In a real healthcare system, convergence of Q-values alone would **not** be sufficient for deployment.

4(c) Final Learned Policy

The policy selects the action having the highest Q-value:

$\pi(s) = \text{argmax}_a Q(s, a)$

Using the illustrative Q-table:

Patient State	Best Action
S0 – High Risk	Medication
S1 – Moderate Risk	Exercise / Medication
S2 – Stable	Diet
S3 – Improved	Monitor
S4 – Deteriorated	Clinical reassessment

Important Clinical Restriction

The Q-table above is only an academic demonstration of reinforcement learning. It must not be used to automatically prescribe medication to real patients.

Ethical Considerations

The assessment specifically requires consideration of patient safety, bias and explainability.

1. Patient Safety

An incorrect treatment recommendation could harm a patient. Therefore, an RL system should operate under strict clinical constraints.

2. Human Oversight

A qualified doctor should approve treatment recommendations.

The AI should provide:

Recommendation → Explanation → Doctor Approval → Treatment

rather than: AI → Automatic Medication

3. Bias

If the training data contain demographic or socioeconomic bias, the RL agent may learn biased policies.

The system should therefore be evaluated across different:

- Age groups
- Sexes
- Geographic populations
- Clinical conditions

4. Explainability

Doctors should be able to understand why the system recommends an action.

Decision Trees are relatively interpretable, while complex RL policies can be more difficult to explain.

5. Privacy

Patient data must be securely stored and processed with appropriate privacy controls.

6. Reward Design

Poorly designed rewards can cause unsafe behavior. For example, rewarding short-term glucose improvement without considering side effects could result in undesirable treatment policies.